BOND PRICING AND THE TIME VALUE OF MONEY

Learning Outcome

- calculate a bond's price given a yield-to-maturity on or between coupon dates

Bond Pricing with a Market Discount Rate

Bond pricing is an application of discounted cash flow analysis that depends on a bond's cash flow features and the rate (or rates) used for discounting. Recall from an earlier lesson that the price of the bond at issuance is the present value of the promised interest and principal cash flows. The market discount rate is used in the time-value-of-money calculation to obtain the present value. The **market discount rate** is the rate of return required by investors given the risk of the bond investment. It is also called the **required yield**, or the **required rate of return**.

The present value (*PV*) calculation for each bond coupon cash flow (*PMT*) that occurs in *t* periods with a market discount rate of *r* per period should be familiar from an earlier time-value-of-money lesson:

$$PV(\text{Bond coupon}) = PMT_t/(1 + r)^t. \qquad 1$$

Equation 2 extends this calculation in a general formula for calculating a bond price (*PV*) given the market discount rate on a coupon date:

$$PV = PMT_1/(1 + r)^1 + PMT_2/(1 + r)^2 + \ldots + (PMT_N + FV_N)/(1 + r)^N, \qquad 2$$

where *FV* is equal to the bond's face value and *N* is the number of periods to maturity.

Recall the Bright Wheels Automotive (BRWA) bond introduced in an earlier lesson. The bond's price was 100% of face or par value because the market discount rate, *r*, and the bond's fixed coupon rate, *PMT*, were equal (at 3.2% per annum, or 1.6% semiannually). We will now consider cases where the bond's coupon rate and the market discount rate are not equal.

EXAMPLE 1

BRWA Par, Discount, and Premium Bond

Period:	1	2	3	4	5	6	7	8	9	10
Cash Flow:	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	101.6
PV of Cash Flow at 1.2% periodic YTM:	1.6	1.6	1.5	1.5	1.5	1.5	1.5	1.5	1.4	90.2
Bond price (sum):	103.75									
PV of Cash Flow at 1.6% periodic YTM:	1.6	1.6	1.5	1.5	1.5	1.5	1.4	1.4	1.4	86.7
Bond price:	100.00									
PV of Cash Flow at 2.0% periodic YTM:	1.6	1.5	1.5	1.5	1.4	1.4	1.4	1.4	1.3	83.3
Bond price:	96.41									

Note: Please refer to the candidate learning ecosystem online for a spreadsheet of these calculations in the downloaded Microsoft Excel workbook.

If the market discount rate rises to 2% semiannually, new bonds to be issued at par must provide coupon rates that are also 2% semiannually. The previously issued BRWA bond pays coupons at a 1.6% rate, so it should have a lower price than a

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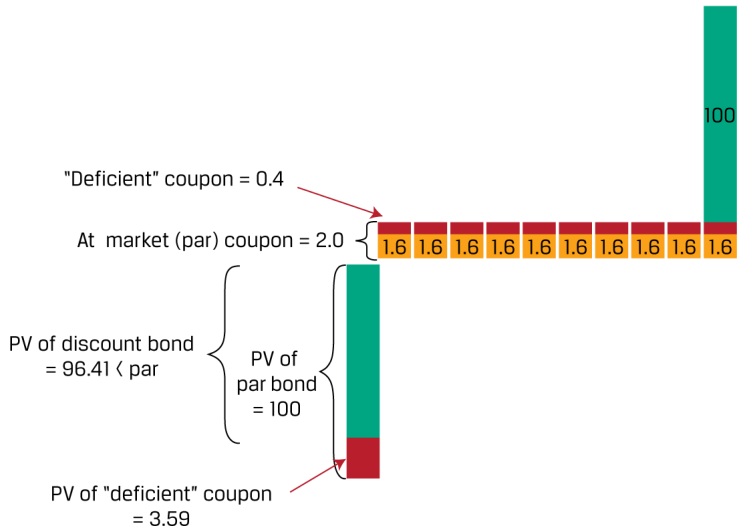
BRWA bond, with its lower coupon, will therefore fall below 100% of par. Example 1 shows that when we discount the BRWA bond cash flows at a market discount rate of 2%, we calculate a price of 96.41. This lower price compensates for a coupon rate below the market discount rate. When a bond's price is less than 100, we say that the bond trades at a discount. (This use of "discount" when referring to the bond price is not connected to the use of "discount" in market discount rates). Exhibit 1 illustrates a bond trading at a discount.

Exhibit 1:

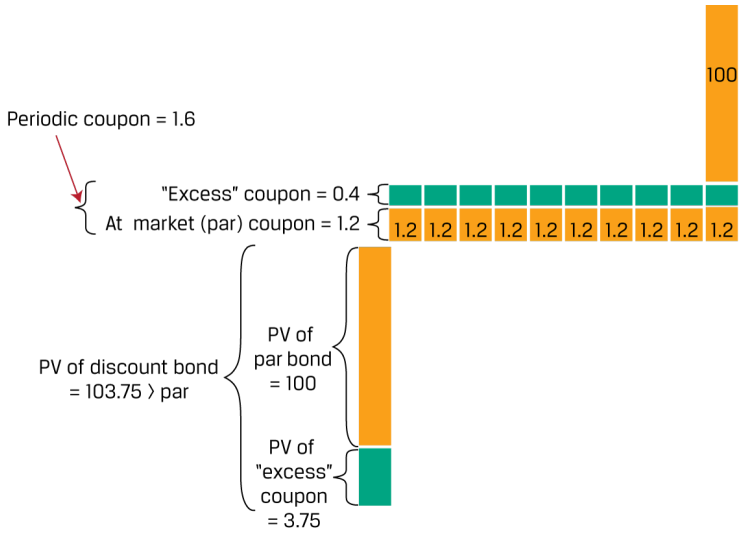
Discount Bond Cash Flows



We now consider how the price of the existing BRWA bond will change if the market discount rate falls to 1.2%. A similar bond (five-year maturity, semi-annual coupon) issued when the five-year market discount rate is 1.2% semiannually will also have a semi-annual coupon rate of 1.2% so that its price will be 100% of par. Because the BRWA bond pays a higher coupon rate of 1.6%, its price should be higher than that of a bond that pays only 1.2%. We could say that the BRWA bond's 1.6% coupon rate is *excessive* relative to the lower 1.2% market discount rate. Therefore, the BRWA bond's price rises as the market discount rate falls. When we discount the cash flows of the BRWA bond at the market discount rate of 1.2% and then add them up, we calculate a bond price equal to 103.73% of par. It has a higher price than the 1.2% coupon bond because it pays a higher coupon. Exhibit 2 illustrates a bond trading at a premium.

Exhibit 2:

Premium Bond Cash Flows



The PV function in Microsoft Excel or Google Sheets may also be used for the price calculation:

$$=PV(RATE,\ NPER,\ PMT,\ FV,\ TYPE),$$

where

rate is the market discount rate per period

nper is the number of periods

pmt is the periodic coupon payment

FV is future or face value

type corresponds to payments made at the end (0) or beginning (1) of each period

The discount, premium, and par relationships are summarized in Exhibit 3.

Exhibit 3:

Fixed-Rate Discount, Premium, and Par Bonds

Bond Price	Price vs. Future Value	Coupon vs. Market Discount Rate
Par	PV = FV	PMT = Market discount rate
Discount	PV < FV	PMT < Market discount rate
Premium	PV > FV	PMT > Market discount rate

The interest rates per period are annualized by multiplying the rate per period by the number of periods in a year. Therefore, an equivalent statement is that the BRWA bond is priced at a premium because its stated annual coupon rate of 3.2% is greater than the stated annual market discount rate of 2.4%. Unless stated otherwise, interest rates are typically quoted as annual rates.

Yield-to-Maturity

If the market price of a bond is known, Equation 2 can be used to calculate its yield-to-maturity. The **yield-to-maturity** is the internal rate of return on the cash flows—the uniform interest rate such that when the future cash flows are discounted at that rate, the sum of the present values equals the price of the bond. It is an *implied* or *observed* single market discount rate.

1. The investor holds the bond to maturity.
2. The issuer makes full coupon and principal payments on scheduled dates. The YTM is the *promised* yield—that is, assuming no issuer default.
3. The investor reinvests all coupon payments at the same YTM.

For example, say we observe that the five-year 3.2% fixed-rate BRWA bond in Example 1 is priced at 108.15. The yield-to-maturity is the solution for the rate, *r*, in the following equation:

$$108.15 = 1.6/(1 + r)^1 + 1.6/(1 + r)^2 + \ldots + 101.6/(1 + r)^{10}.$$

The Microsoft Excel or Google Sheets YIELD function may be used for bonds that pay periodic interest:

`=YIELD(SETTLEMENT, MATURITY, RATE, PR, REDEMPTION, FREQUENCY, [BASIS])` where

settlement is the settlement date entered using the DATE function

maturity is the maturity date entered using the DATE function

rate is the semi-annual (or periodic) coupon

pr is the price per 100 face value

redemption is the future value at maturity

frequency is the number of coupons per year

[basis] is the day-count convention, typically 0 for US bonds (30/360 day count).

If we assume the BRWA bond settles and matures on 15 October 2025 and 15 October 2030, respectively, the appropriate entries are DATE(2025,10,15) and DATE(2030,10,15). The other entries are *rate* = 1.6%, *pr* = 108.15, *redemption* = 100, *frequency* = 2, and *basis* = 0, as follows:

=YIELD(DATE(2025,10,15),DATE(2030,10,15),1.6%,108.15,100,2,0)

Given a price of 108.15, the YIELD function calculates the annual yield-to- maturity for the BRWA bond to be 1.50%. Additionally, since the yield-to-maturity is an internal rate of return, we can also use the IRR function in Microsoft Excel or Google Sheets to get the same result.

Yield-to-maturity, commonly referred to as “yield,” is often used interchangeably with the market discount rate or required yield and in place of discussing bond prices. For example, market participants will simply say “yields are rising” rather than “market discount rates are rising” or “bond prices are falling.”

Bonds with Negative Yields-to-Maturity

The yield-to-maturity on a bond may be positive, zero, or negative. Yields on many sovereign bonds fell below zero following accommodative monetary policies introduced by central banks starting in 2012 in response to below-target inflation rates. Bonds with a negative yield-to-maturity include those issued at higher yields in the past that have experienced significant price appreciation and newly issued zero-coupon sovereign government bonds priced at a premium to par value. Bonds are generally not issued with negative coupons.

In 2020, China issued its first ever negative-yielding sovereign bond denominated in euros, with terms shown in Exhibit 4.

Exhibit 4: People’s Republic of China Eurobond

Issuer:	People's Republic of China
Settlement Date:	November 25, 2020
Maturity Date:	November 25, 2025
Principal Amount:	EUR 750 million
Interest:	Zero coupon (ACT/ACT day count)
Issuance Price:	100.763
Issuance Spread:	+30 bps p.a. plus Mid Swaps
Issuance and Trading:	Eurobond Listed on Multiple Exchanges Euroclear/ Clearstream
Seniority:	Senior Unsecured
Exchanges:	Luxembourg

The Excel YIELD function cannot not be used for zero-coupon bonds. Instead, we solve for the annualized yield-to-maturity using Equation 1 for the single cash flow of par (*PMT* = 100) after five years (*t* = 5) at the original issuance price (*PV* = 100.763):

$$PV = \frac{PMT}{(1+r)^t}.$$

$$100.763 = \frac{100}{(1+r)^5}.$$

$$r = \left(\frac{100}{100.763}\right)^{\frac{1}{5}} - 1.$$

$$r = -0.15\%.$$

The *Financial Times* reported high demand for this negative-yielding Chinese bond since its yield was more than 50 bps above that of European sovereign bonds—the five-year German bund was at approximately –0.74% around the same time —and more generally because many investors sought portfolio exposure to China amid a relative scarcity of euro-denominated Chinese bonds at the time.

Flat Price, Accrued Interest, and the Full Price

When a bond is priced between coupon payment dates, its price has two components: the **flat price** (PV^{Flat}) and the **accrued interest** (*AI*). The sum of the parts is the **full price** (PV^{Full}), which also is known as the invoice or "dirty" price.

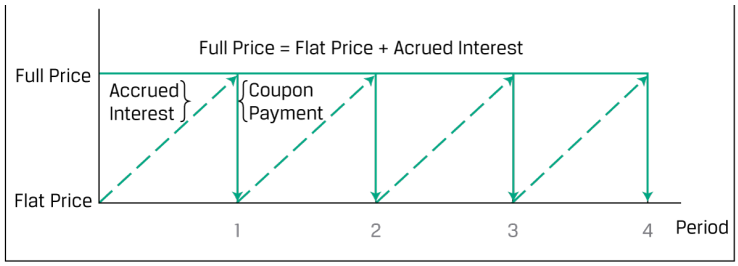
$$PV^{Full} = PV^{Flat} + AI. \quad (3)$$

The flat price, which is the full price minus the accrued interest, is also called the quoted or "clean" price and is the type of price usually quoted by bond dealers. If a trade takes place, the accrued interest is added to the flat price to obtain the full price paid by the buyer and received by the seller on the **trade settlement date**. The trade settles when the bond buyer pays and the seller delivers the security.

The flat price is used for quotations to avoid misleading investors about a bond's market price trend. If full prices were quoted by dealers, investors would see price rises each day even if the YTM did not change due to the accrual of interest. Once a coupon payment is made, the quoted price would drop significantly, as shown in Exhibit 5.

Exhibit 5:

Flat Price, Accrued Interest, and Full Price



▶ Expand / collapse Extended Description

The accrued interest is the portion of the next coupon payment owed to the seller of a bond, because although the seller held the bond for a partial coupon period, the full coupon will be received by the buyer. To calculate accrued interest, we determine the fractional amount by counting the days in the period. If the coupon period has *T* days between payment dates and *t* days have passed since the last payment, the accrued interest is calculated using Equation 4:

$$AI = \frac{t}{T} \times PMT, \qquad 4$$

where

t = number of days from the prior coupon payment to the settlement date

T = number of days in the coupon period

t/*T* = fraction of coupon period that has passed since the prior payment

PMT = coupon payment per period

Note that the accrued interest portion of the full price does not depend on the yield-to-maturity. Therefore, it is only the flat price that is affected by a change in interest rates.

There are multiple approaches for day counting (or "day counts"). The day count is specified by two parameters: how days are counted within a given period and the number of days assumed per period. Two common day count conventions are 30/360 and actual/actual. The 30/360 day count assumes each month has exactly 30 days (although most don't) and each year has exactly 360 days (although none do). In contrast, the actual/actual day count convention assumes the actual number of days in each month and the actual number of days in a year.

EXAMPLE 2

Actual/Actual vs. 30/360 Day Count

Consider two otherwise identical bonds that pay 4.375% semiannual interest on 15 May and 15 November of each year. One is a government bond with an actual/actual day count and the other is a corporate bond with a 30/360 day count. The actual/actual method uses the actual number of days, including weekends, holidays, and leap days. The 30/360 method assumes 30 days in a month and 360 days in each year.

Calculate accrued interest for settlement on 27 June for both bonds.

Accrued interest for the actual/actual bond is the actual number of days between 15 May and 27 June (*t* = 43 days) divided by the actual number of days between 15 May and 15 November (*T* = 184 days), times the coupon payment.

For the 30/360 bond, we 42 days between 15 May and 27 June: 15 days between 15 May and 30 May and 27 days between 1 June and 27 June (31 May is not counted in the 30/360 convention). There are assumed to be 180 days in the six-month period between 15 May and 15 November.

Days (t)	43	42
Period (T)	184	180
t/T	0.233696	0.233333

Accrued interest is calculated as follows:

Actual/actual bond: AI = (43/184) × (4.375/2)

= 0.511209 per 100 of par value.

30/360 bond: AI = (42/180) × (4.375/2)

= 0.510417 per 100 of par value.

The full price of a fixed-rate bond between coupon payments given the market discount rate per period (r) is the present value of future cash flows as of the trade settlement date, as shown in Equation 5:

$$PV^{Full} = \frac{PMT}{(1+r)^{1-t/T}} + \frac{PMT}{(1+r)^{2-t/T}} + \cdots + \frac{PMT+FV}{(1+r)^{N-t/T}}. \quad (5)$$

This is very similar to Equation 2. The difference is that the next coupon payment (PMT) is discounted for the remainder of the coupon period, which is $1 - t/T$. The second coupon payment is discounted for that fraction plus another full period, $2 - t/T$.

Equation 5 is simplified by multiplying both the numerator and denominator by the expression $(1 + r)^{t/T}$. The result is Equation 6:

$$\begin{aligned} PV^{Full} &= \left[\frac{PMT}{(1+r)^{1}} + \frac{PMT}{(1+r)^{2}} + \cdots + \frac{PMT+FV}{(1+r)^{N}}\right] \times (1+r)^{t/T} \\ &= PV \times (1+r)^{t/T}. \end{aligned} \quad (6)$$

An advantage of Equation 6 is that the expression in the brackets, PV, is easily obtained using the Excel YIELD function because there are N evenly spaced periods. PV here is the present value of future cash flows as of the last coupon date; it is identical to Equation 2 but is not the same as PV^{Flat}.

The next two examples demonstrate the calculation of a bond's flat price, accrued interest, and full price.

EXAMPLE 3

BRWA Bond—Flat and Full Price

Calculate the flat price and full price of the 3.2% fixed BRWA bond 90 days after the first coupon payment.

First, we calculate the bond's present value on the coupon date that begins the period during which settlement occurs, using Equation 2 or the PV function in Microsoft Excel or Google Sheets.

```
=–PV(RATE, NPER, PMT, [FV], TYPE)
```

=-PV(0.02,9,1.60,100,0)

= 96.735

Next, we calculate the full price by counting the days using the specified day count convention and using Equation 6.

Number of days from beginning of period until settlement, t = 90

Number of days in the period containing settlement, T = 180

$$PV^{Full} = PV \times (1+r)^{t/T}$$

$$PV^{Full} = 96.735 \times (1+0.02)^{90/180}$$

$$AI = \frac{t}{T} \times PMT$$

$$AI = \frac{90}{180} \times 1.60$$

$$AI = 0.80$$

Finally, we calculate the flat price using Equation 3 by subtracting the accrued interest from the full price.

$$PV^{Full} = PV^{Flat} + AI$$

$$PV^{Flat} = PV^{Full} - AI$$

$$PV^{Flat} = 97.698 - 0.80$$

$$PV^{Flat} = 96.898$$

Note: Quantities in the calculation are not rounded, though they are shown with three decimal places for presentation purposes. Solutions are rounded to three decimal places.

EXAMPLE 4

Romania Bond—Flat and Full Price

The 4.625% EUR annual, actual/actual fixed-coupon bond issued by Romania maturing 3 April 2049 was introduced in an earlier lesson. Calculate the full price (PV^{Full}), the accrued interest (AI), and the flat price (PV^{Flat}) for settlement on 15 December 2031 assuming an annual yield-to-maturity on that date of 3.50%.

Solution

Step 1: Calculate PV as of the beginning of the current coupon period for the settlement date:

```
=–PV(RATE, NPER, PMT, [FV], TYPE)
```

=-PV(0.035,18,4.625,100,0)

= 114.838

Step 2: Calculate the actual number of days (t):

Days	Months
15	December
27	April
30	June, September, November
31	May, July, August, October
256	**Total**

Step 3: Calculate the actual number of days in the coupon period (T):

Date at beginning of period	4/3/2031
Date at end of period	4/3/2032
Actual number of days in period	**366**

Notice that 2032 is a leap year so it has an extra day.

Step 4: Use Equation 6 to solve for PV^{Full}:

$$PV^{Full} = PV \times (1 + r)^{\frac{t}{T}}.$$

$$PV^{Full} = 114.838 \times (1 + 0.035)^{\frac{256}{366}}.$$

$$PV^{Full} = 117.635.$$

$$AI = \frac{\tau}{T} \times PMT.$$

$$AI = \frac{256}{366} \times 4.625.$$

$$AI = 3.235.$$

Step 6: Use Equation 3 to solve for PV^{Flat}:

$$PV^{Full} = PV^{Flat} + AI.$$

$$PV^{Flat} = PV^{Full} - AI.$$

$$PV^{Flat} = 117.635 - 3.235.$$

$$PV^{Flat} = 114.400.$$

Note: Quantities in the calculation are not rounded, though they are shown with three decimal places for presentation purposes. Solutions are rounded to three decimal places.

Knowledge Check

1. Define yield-to-maturity.

Hide Solution

Solution:

Yield-to-maturity is the internal rate of return on a bond's cash flows—the uniform interest rate such that when the future cash flows are discounted at that rate, the sum of the present values equals the price of the bond.

2. A non-callable fixed-coupon bond with a maturity in 10 years and a face value of 100 is issued with an annual coupon rate of 2.8% when the applicable market discount rate is also 2.8%. The price of this bond at issuance is:

A. less than 100.
B. exactly 100.
C. more than 100.

Hide Solution

Solution:

B is correct. When a bond is issued with a coupon rate that equals the market discount rate, it is priced at par.

3. A non-callable fixed-coupon bond with a maturity in 10 years and a face value of 100 has an annual coupon rate of 2.8%, while the market discount rate is 4.0%. The price of this bond at issuance is:

A. less than 100.
B. exactly 100.
C. more than 100.

Hide Solution

Solution:

A is correct. A bond with a coupon rate lower than the market discount rate will trade at a discount to par, to compensate for its "deficient" coupons relative to the market rate.

4. A sovereign bond that matures on 1 January 2030 pays an annual coupon of 1.2% and has a face value of 100. For a trade settlement date of 1 January 2020, the

Hide Solution

Solution:

–1.390%. We use the YIELD function in Microsoft Excel or Google Sheets:

```
=YIELD(SETTLEMENT, MATURITY, RATE, PR, REDEMPTION, FREQUENCY,[BASIS])
```

=YIELD(DATE(2020,1,1),DATE(2030,1,1),0.012,128,100,1)

= -1.390%

5. A bond with a face value of 100 pays an annual coupon of 12% and matures on Ad14 August 2007. It uses the actual/actual day count convention. For a settlement date of 23 December 2002 and an annual YTM of 9.75%, calculate the full price, accrued interest, and flat price. Assume an actual/actual day count.

Hide Solution

Solution:

Step 1 Calculate PV as of the beginning of the current coupon period for the settlement date:

```
=–PV(RATE, NPER, PMT, [FV], TYPE)
```

=-PV(0.0975,5,12,100,0)

= 108.584

Step 2 Calculate the actual number of days (*t*):

Month	Days
August	17
September	30
October	31
November	30
December	23
Total	**131**

Step 3 Calculate the actual number of days in the coupon period (*T*):

Date at beginning of period	8/14/2002
Date at end of period	8/14/2003
Actual number of days in period	**365**

Step 4 Use Equation 6 to solve for PV^{Full}:

$$PV^{Full} = PV \times (1+r)^{\frac{t}{T}}.$$

$$PV^{Full} = 108.584 \times (1+0.0975)^{\frac{131}{365}}.$$

$$PV^{Full} = 112.271.$$

Step 5 Use Equation 4 to solve for accrued interest:

$$AI = \frac{t}{T} \times PMT.$$

$$AI = \frac{131}{365} \times 12.$$

$$AI = 4.307.$$

Step 6 Use Equation 3 to solve for PV^{Flat}:

$$PV^{Full} = PV^{Flat} + AI.$$

$$PV^{Flat} = 107.964.$$

Note: Quantities in the calculation are not rounded, though they are shown with three decimal places for presentation purposes. Solutions are rounded to three decimal places.

6. Perform the same calculations for the same bond in Question 5, but use a 30/360 day count convention.

Hide Solution

Solution:

Step 1 Calculate PV as of the beginning of the current coupon period for the settlement date:

=–PV(RATE, NPER, PMT, [FV], TYPE)

=-PV(0.0975,5,12,100,0)

= 108.584

Step 2 Calculate the number of days (*t*):

Month	Days
August	16
September	30
October	30
November	30
December	23
Total	**129**

Step 3 Assume the number of days in the coupon period (*T*) is 360.

Step 4 Use Equation 6 to solve for PV^{Full}:

$$PV^{Full} = PV \times (1 + r)^{\frac{t}{T}}.$$

$$PV^{Full} = 108.584 \times (1 + 0.0975)^{\frac{129}{360}}.$$

$$PV^{Full} = 112.265.$$

Step 5 Use Equation 4 to solve for accrued interest:

$$AI = \frac{t}{T} \times PMT.$$

$$AI = \frac{129}{360} \times 12.$$

$$AI = 4.300.$$

Step 6 Use Equation 3 to solve for PV^{Flat}:

$$PV^{Full} = PV^{Flat} + AI.$$

$$PV^{Flat} = PV^{Full} - AI.$$

$$PV^{Flat} = 112.265 - 4.300.$$

$$PV^{Flat} = 107.965.$$

Note: Quantities in the calculation are not rounded, though they are shown with three decimal places for presentation purposes. Solutions are rounded to three decimal places.

Discussion

What do you want to discuss... (required)

JA

are you suggesting we can use excel on the exam?

Created 2 months ago by Johnathon Adam Jones — 0 replies | Last Activity: 2 months ago

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SG

question 5

the strokes for calci are

step 1- press 2nd then press 9 i.e the bond function (arrow down)

step 2- it asks for std i.e settlement date so write 23.1202 (23 dec 2002) (arrow down)

step 3- it asks cpn i.e coupon so it is 12 (arrow down)

step 4- it asks rdt i,e redemption date so write 14.0807 (arrow down)

step 5- it asks rv it is 100 (arrow down)

step 6- it asks for *act/act* or *360* so for question 5 it is act for question 6 it is 360 so to change u can press 2nd then press enter it changes (arrow down)

step 7- it asks *1/y* anual or *2/y* semianual so according to question u change to change press 2nd then press enter (arrow down)

step 8- it asks yield put the yield given in question donot change even if it mentions semi annual (arrow down)

step 9- here it says pri then press cpt it is the flat price to get full price press arrow down it gives accrued int so add pri and ai u get the full price

Created 2 months ago by SAURAV GOGOI — 0 replies | Last Activity: 2 months ago

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YG

In example 3, is it just me or was the market discount rate not clearly communicated? Where is it mentioned that we should take 2% semianually?

Created 4 months ago by Yanis Guessoum — 1 reply | Last Activity: 3 months ago

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ZA

hi, in example 4 my calculator gives a flat price 107.964 and accrued interest 4.307 which is correct. showing excel calculation really confuses

Created 5 months ago by Zamina Akbarova — 2 replies | Last Activity: 4 months ago

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JR

When I calculate Example 4 using the calculator I am getting 116.9734 while in excel it shows 114.84. I am not sure if I am wrong.

Created 6 months ago by Johanne Roshini Robert William — 2 replies | Last Activity: 4 months ago

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MN

why doesn't the curriculum include a lot of explanations on the use of the calculator as opposed to excel? I am asking so because our exams do not allow the use of excel

Created 4 months ago by Medson Ntheye Nyirenda — 0 replies | Last Activity: 4 months ago

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IR

Does anyone know how is calculating the Number of Periods (NPER) in both examples? Many thanks for your time!

Created 4 months ago by Ibon Rodríguez — 1 reply | Last Activity: 4 months ago

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AZ

" Q. A three-year, semiannual-pay bond with a $100 par value and a 5% coupon rate is purchased for $108. One year later, if the yield to maturity has decreased by 100 basis points, the change in the value of this bond is closest to "

This question was placed in this module even though the concept (impact of yield change on bond price) is only introduced later in the curriculum. The timing feels premature for this section

Created 4 months ago by Abdulaziz Zahid — 0 replies | Last Activity: 4 months ago

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JT

why tell us how to solve using excel..? and not the tool we will be using on the exam?

BA Please may someone show me the inputs they used to solve for PV in example 3 using a calculator

Created 6 months ago by Bradley Alan Palmer 1 reply | Last Activity: 6 months ago

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